

Dr Viridiana Lizardo Briseño

Postdoctoral Researcher (PINCC – FUNAM)
Entomology | Evolutionary biogeography | Collections management
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Research interests: Systematics and curation of Coleoptera, evolutionary biogeography and macroecology, with emphasis on the use of biological collections and geospatial data in evolutionary and conservation studies.

Education

PhD in Biological Sciences (2021–2025). Department of Evolutionary Biology, Faculty of Sciences, UNAM.

Thesis: *Conservación filogenética de especies en la Zona de Transición Mexicana*. [Phylogenetic conservation of species in the Mexican Transition Zone]

MSc in Biological Sciences (2013–2015). Department of Zoology, Institute of Biology, UNAM.

Thesis: *Distribución potencial de Phanaeini (Coleoptera: Scarabaeidae) en México y su posible modificación por el cambio de uso del suelo*. [Potential distribution of Phanaeini (Coleoptera: Scarabaeidae) in Mexico and its possible modification by land-use change]

BSc in Biology (2007–2011). Faculty of Sciences, UNAM.

Thesis: *Análisis de la comunidad de escarabajos coprófagos de Villa de Cos, Zacatecas, México*. [Analysis of the dung beetle community of Villa de Cos, Zacatecas, Mexico]

Skills

Entomological work: Taxonomic identification, collection digitisation, specimen mounting. Entomological sampling, survey design and diversity analysis.

Biological database management: Data cleaning and analysis, taxonomic harmonisation, georeferencing.

Spatial analysis: Species distribution modelling; analysis of diversity patterns (taxonomic, phylogenetic and functional).

Phylogenetic comparative analysis: Use of phylogenetic data in community studies and application of comparative methods.

Programming and statistical analysis: Proficient in R; basic Python for data processing; advanced QGIS.

Academic and professional experience

Research experience

2025–present. **Project «Fortalecimiento de capacidades para la toma de decisiones en cambio climático y actualización de las contribuciones nacionalmente determinadas»**. [Capacity building for climate change decision-making and updating of the Nationally Determined Contributions] Climate Change Research Programme (PINCC), UNAM, and Institute of Biology, UNAM.

2026. **External consultant, Country Climate and Development Report (CCDR)**. World Bank, Agriculture and Food.

2023. **Species distribution modelling and spatial analysis of species richness**. Universidad Autónoma de Tlaxcala y Centro Tlaxcala de Biología de la Conducta.

2014. **Research stay on species distribution modelling**. Instituto de Ecología, A.C. (INECOL), Xalapa, Veracruz, Mexico.

Collections and biological data management

2015. **Counting, sorting and cleaning of specimens and species identification of Scarabaeidae**. Oxford University Museum of Natural History, Operation Wallacea project.

2013–2015. **Distribution modelling of Phanaeus species using GARP, R and QGIS, and database compilation**. CONABIO project JM032 («Distribución potencial en taxones hiperdiversos y poco conocidos (Insecta: Coleoptera: Scarabaeoidea): estimando la localización de nuevos taxones y el efecto de los cambios en los usos del suelo»).

2009–2010. **Specimen maintenance in an insect collection**. Social service placement, Colección Nacional de Insectos, Instituto de Biología, UNAM.

Teaching

2024–2025. **Geographic Information Systems**. Universidad Autónoma de Zacatecas, Unidad Académica de Agronomía, Ingeniería en Agronomía y Zootecnia.

2024. **Vertebrates**. Centro de Actualización del Magisterio, Licenciatura en Enseñanza de la Biología.

2024. **Multicellular autotrophic organisms**. Centro de Actualización del Magisterio, Licenciatura en Enseñanza de la Biología.

2023. **Postgraduate course: Biostatistics in R**. Universidad Autónoma de Tlaxcala, Posgrado en Ciencias Biológicas.

Peer-reviewed articles

- Lizardo, V.**, & Ruggiero, A. (2026). Biogeographic boundaries and angiosperm phylogenetic endemism: highlighting the evolutionary complexity of the South American Transition Zone. *Journal of Biogeography*.
- Lizardo, V.**, Ruggiero, A., & Morrone, J. J. (2025). Patrones de riqueza de especies y conservadurismo filogenético del nicho ecológico en la Zona de Transición Mexicana: evidencia y herramientas para su estudio. *Revista Mexicana de Biodiversidad*, 96, e965347.
- Lizardo, V.**, Escobar, F., Martínez-Meyer, E., & Morrone, J. J. (2024a). Adaptive shifts in Phanaeini dung beetles of the Mexican plateau cenocron in the Mexican transition zone. *Zoologica Scripta*. <https://doi.org/10.1111/zsc.12656>
- Lizardo, V.**, García Trejo, E. A., & Morrone, J. J. (2024b). Niche conservatism and convergence in birds of three cenocrons in the Mexican Transition Zone. *PeerJ*, 12, e16664. <https://doi.org/10.7717/peerj.16664>
- Moctezuma, V., **Lizardo, V.**, Arias-Del Razo, I., & Ramírez-Ponce, A. (2024). Overcoming the Wallacean shortfall in sky-islands of central Mexico: the case of copro-necrophagous beetles and two national parks. *Journal of Insect Conservation*. <https://doi.org/10.1007/s10841-024-00598-9>
- Moctezuma, V., Hernández, B., Sánchez-Huerta, J. L., **Lizardo, V.**, Quiroz-Rocha, G. A., & Navarrete-Heredia, J. L. (2023). Two new species from the Mexican Pacific Slope and new distributional records of *Onthophagus* Latreille, 1802 (Coleoptera, Scarabaeidae). *Journal of Asia-Pacific Entomology*, 26(1), 102015. <https://doi.org/10.1016/j.aspen.2022.102015>
- Lizardo, V.**, Moctezuma, V., & Escobar, F. (2022). Distribution, regionalization, and diversity of the dung beetle genus *Phanaeus* MacLeay (Coleoptera: Scarabaeidae) using species distribution models. *Zootaxa*, 5213(5), 546–568. <https://doi.org/10.11646/zootaxa.5213.5.4>
- Moctezuma, V., Halffter, G., & **Lizardo, V.** (2021). The *Phanaeus tridens* species group (Coleoptera: Scarabaeoidea): a dung beetle group with genital morphological stasis but a changing ecological niche. *Acta Entomologica Musei Nationalis Pragae*, 447–482. <https://doi.org/10.37520/aemnp.2021.025>
- Lizardo, V.**, Escobar, F., & Rojas-Soto, O. (2017). Diversity and distribution of Phanaeini (Coleoptera: Scarabaeidae: Scarabaeinae) in Mexico. *Zootaxa*, 4358(2), 271. <https://doi.org/10.11646/zootaxa.4358.2.3>
- Lizardo, V.**, & Castellanos-Vargas, I. (2016). Dung beetle community response to vegetation type and season in an arid zone of the Mexican Plateau. *Southwestern Entomologist*, 41(2), 441–452. <https://doi.org/10.3958/059.041.0215>

Conference participation

- LIX Congreso Nacional de Entomología. 2024. San Luis Potosí, Mexico.
- XIII Reunión Latinoamericana de Scarabaeidología. 2022. Guadalajara, Jalisco, Mexico.
- British Ecological Society Macrofest. 2022. Sheffield, UK.
- Oxford University Mexican Society's Talk About. 2019. Oxford, UK.
- 5.º Congreso Mexicano de Ecología. 2015. San Luis Potosí, Mexico.
- X Reunión Latinoamericana de Scarabaeidología. 2014. Villa de Leyva, Boyacá, Colombia.
- 3.er Congreso Mexicano de Ecología. 2011. Boca del Río, Veracruz, Mexico.

Science communication and outreach

Public lectures and talks

- 2025-04. **Festival María Sibylla Merian – charla «Ciencia y arte»**. [Science and art] Zacatecas, Mexico. Talk on science communication through literature and art, drawing on the editorial experience of *Un librito de bichos*.
- 2025-02-28. **Notas sobre la evolución biológica de la menstruación**. [Notes on the biological evolution of menstruation] Zacatecas, Mexico. Public lecture on the evolution of menstruation, historical biases in its scientific study and current perspectives on the female body in biological research.
- 2022-07-11. **Apocalipsis de los insectos**. [The insect apocalypse] Zigzag, Museo Interactivo de Ciencia y Tecnología, Zacatecas, Mexico. Online lecture on the decline of insect populations.
- 2021-03-25. **Escarabajos coprófagos, trabajo sucio para monitorear la diversidad**. [Dung beetles: dirty work for monitoring diversity] First lecture series «Evolución, sociedad y ambiente», Colegio de Ciencias y Humanidades, UNAM, Plantel Oriente, Mexico City. Online lecture on the ecological relevance of dung beetles as biodiversity indicators.

Popular science articles

2024. **Lizardo, V.** Un pinacate, ¿apestaría igual si se llamara diferente? *Puntos Suspensivos*, suplemento «El Mechero».

2023. López, S., Vega, V., Núñez, A., Cifuentes, P., **Lizardo, V.**, González, M. & Aquino, A. Mexicanas, nombrando, describiendo y clasificando la biodiversidad. *Ciencias y Más*, 3(12): 5–9.

Editorial and outreach activities

- 2024-05-16. **Launch of *Un librito de bichos***. Complejo Cultural Los Pinos, Mexico City. Public launch of a children’s science book, with reading-mediation and environmental education activities.
- 2020-10-30. **Biodiversidad para niñas, niños y adolescentes**. [Biodiversity for children and teenagers] 6th Fandango de Arte Infantil «Gachita Amador», Instituto Zacatecano de Cultura Ramón López Velarde, Zacatecas. Series of educational videos with background content and activities for studying biodiversity.
- 2018-03-04. **Cooking class: Traditional Mexican food**. Oxford University Mexican Society, Oxford, UK. Cultural workshop with a historical introduction to Mexican cuisine.

Interdisciplinary art–science projects

- 18 Sep – 26 Oct 2025. ***La dignidad del insecto: vida y muerte de lo imperceptible***. [The dignity of the insect: life and death of the imperceptible] Interdisciplinary project (playwriting, podcast and field diary) integrating entomological content and environmental reflection; season at Teatro El Galeón, Centro Cultural del Bosque, Mexico City. Co-produced by Teatro UNAM and the Instituto Nacional de Bellas Artes y Literatura (INBAL).

Awards, grants and distinctions

Year	Institution	Award / grant
2025	Red Mexicana de Biodiversidad	«Dra. Valeria Souza Saldívar» Award for Best Doctoral Thesis, presented at the 1st Mexican Congress of Evolution.
2025	UNAM	Doctoral thesis awarded honourable mention.
2025	UNAM	Timely Graduation Award (PhD).
2025	INBAL	Grant from the 2025 Theatre Programme for Children, Young People and Adults, for the project <i>La dignidad del insecto: vida y muerte de lo imperceptible</i> .
2020–2025	SECIHTI	National Postgraduate Scholarship (PhD).
2020	Instituto Zacatecano de Cultura «Ramón López Velarde»	Honourable mention, «Publicación Literaria Infantil y Juvenil» (children’s and young adult literature) category.
2019	Secretaría de Finanzas del Estado de Zacatecas	Finalist, «10 mejores ideas de negocio» (top 10 business ideas) competition, Startup Zacatecas programme.
2015	UNAM	Timely Graduation Award (MSc).
2013–2015	CONACyT	National Postgraduate Scholarship (MSc).
2014	UNAM	Postgraduate Support Programme (PAEP) grant for a one-month research stay at the Instituto de Ecología, A.C. (INECOL).
2010	UNAM	Undergraduate Academic Strengthening Programme (PFEL).

Literary work

Playwriting

2025. **Co-author. *La dignidad del insecto: vida y muerte de lo imperceptible***. [PUBLISHER]. ISBN 978-970-96984-1-1. Interdisciplinary project combining theatre, podcast and field diary to highlight the ecological relevance of insects in the face of the environmental crisis; season at Teatro El Galeón, Centro Cultural del Bosque, Mexico City (co-produced by INBAL and Teatro UNAM).

Children's books

- [AÑO]. ***Es natural cambiar***. [It is natural to change] [PUBLISHER]. ISBN 978-607-26868-3-0.
2023. ***Un librito de bichos***. [A little book of bugs] Amonite. An illustrated introduction in verse to the insect diversity of the Mexican Plateau region, for children.

Short fiction

- 2024-02-26. **«Servicio de mudanza»**. [Removal service] *Puntos Suspensivos* («El Mechero» supplement), Zacatecas, Mexico.

Academic references

Available upon request.